Homeowners Pricing

A proposal from Alpha Actuarial

Introduction



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Overview

The problem:

- The current pricing overprices standard risks while underpricing mitigated risks.
- Competitors offer flat discounts that don’t reflect actual loss experience across key perils.
- Risks of losing low risk, profitable policyholders

Our solution:

- Implement lower, targeted discount factors for policyholders with protective features.
- Rebalance base rates to preserve overall portfolio revenue and ensure solvency.
- Capture high-margin, low risk market share from competitors.

Assumptions



Data

- Historical data is representative of future Claims
- Claims are fully developed (No IBNR)



Pricing

- Company expenses are spread proportionally across all policyholders
- Perils are independent and not every peril is given a discount
- Overall premium level is preserved after rebasing

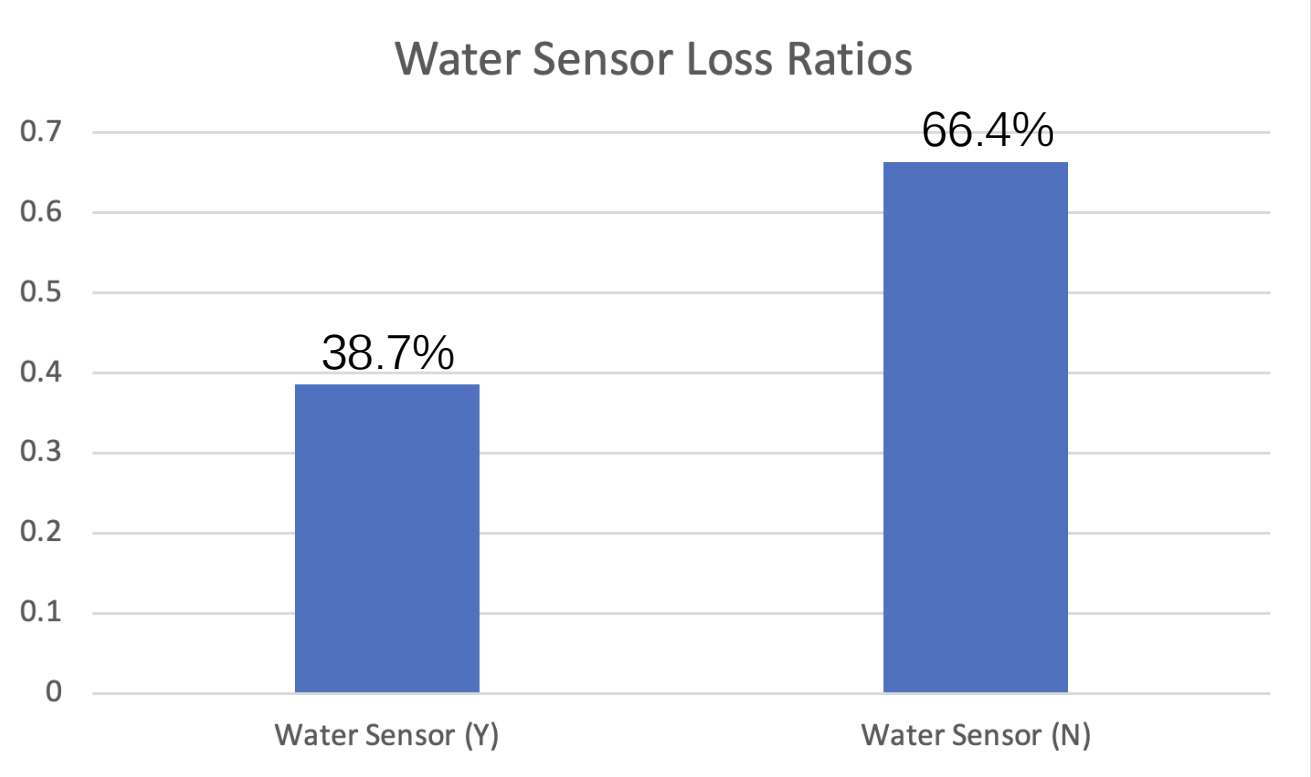


Market

- Competitor discounts are representative of current market pricing

Loss Ratio Analysis

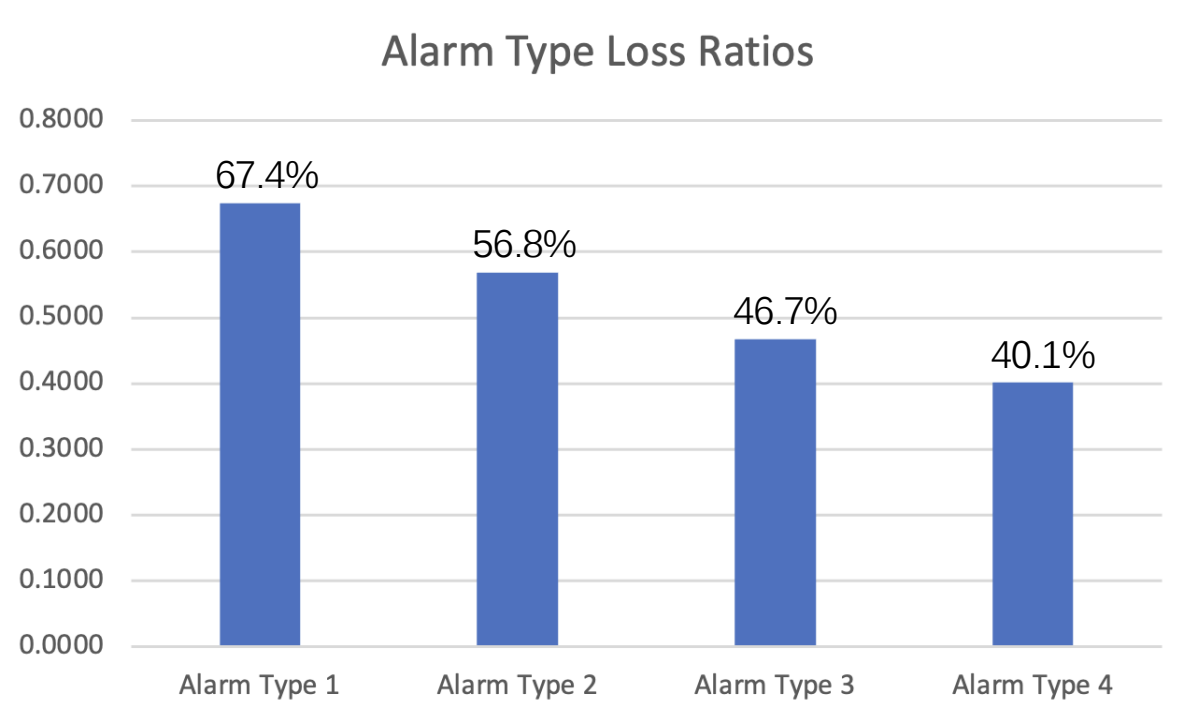
$$\textbf{Loss Ratio} = \frac{\text{Losses}}{\text{Premium}}$$



Using Water Data only

Loss Ratio Analysis

Using Theft Data only



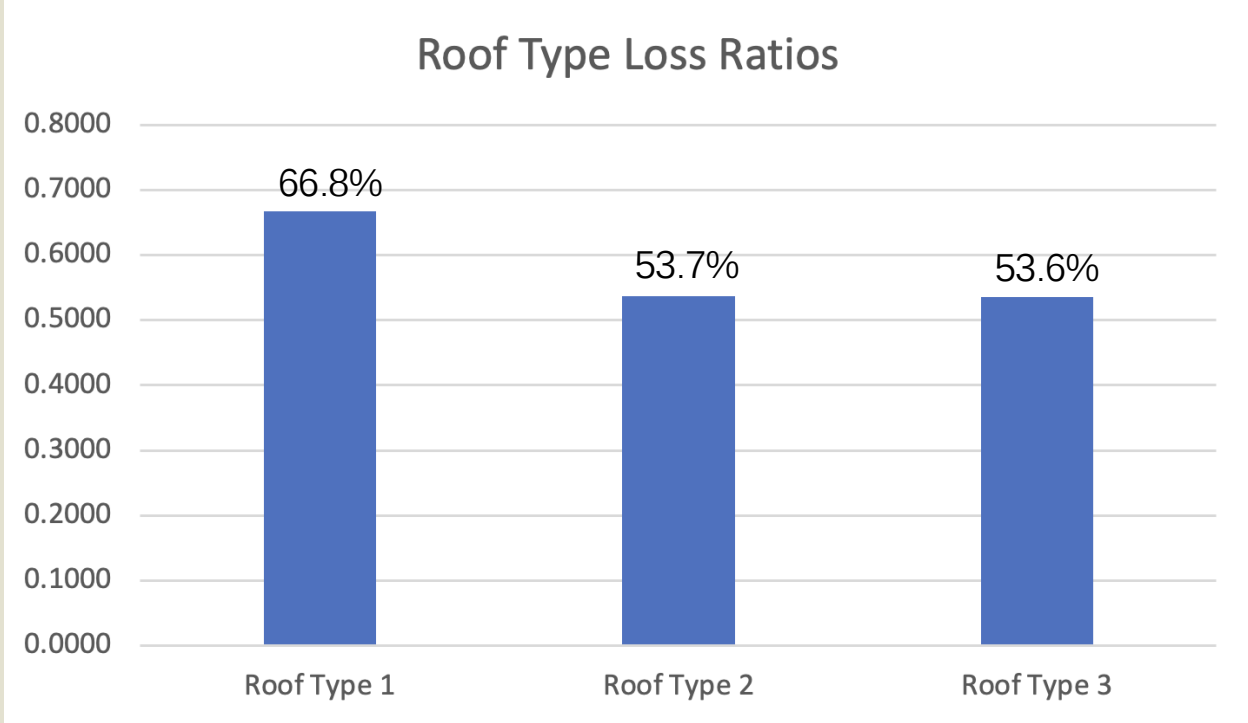
Alarm 1 = None

Alarm 2 = Local Alarm

Alarm 3 = High Tech Local Alarm

Alarm 4 = Monitored High Tech Alarm

Using Wind/Hail Data only



Roof 1 = Composite Shingle

Roof 2 = Hail Resistant Shingle

Roof 3 = Solar Panel over Shingle

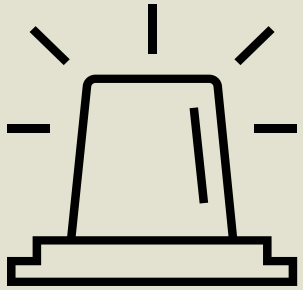
Loss Ratio Key Findings

Water Sensor



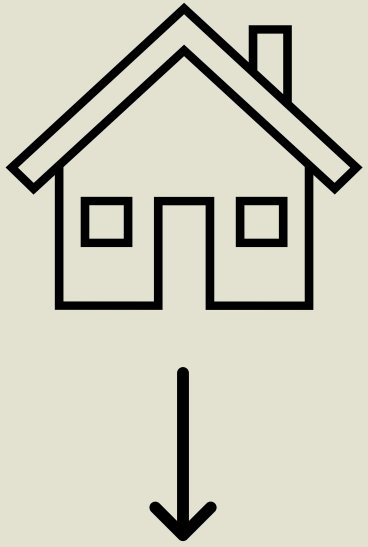
Largest reduction in LR

Alarm



Higher protection = lower LR

Roof



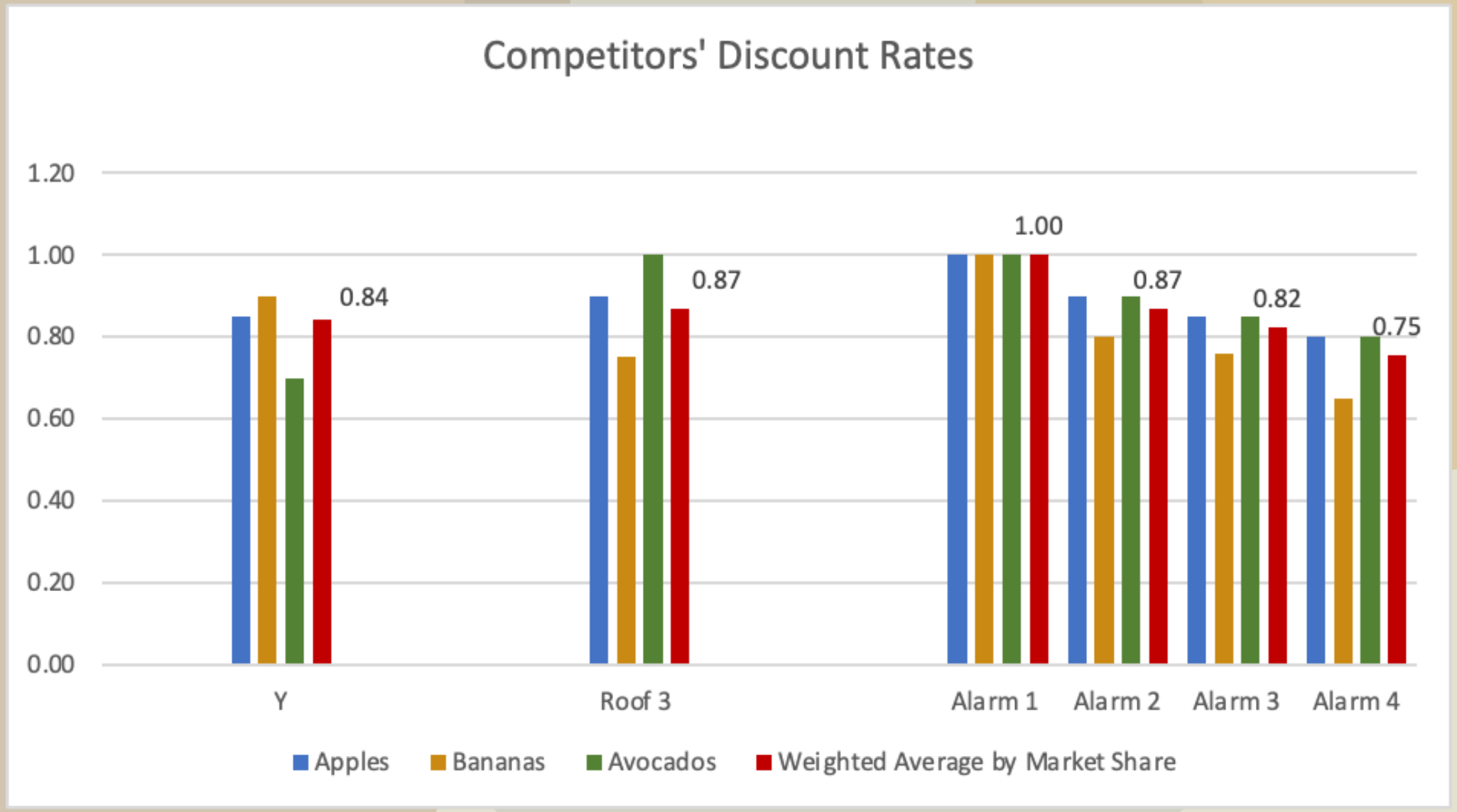
Type 2 ≈ Type 3

Credibility

Group	# of Claims
Water	
Y	722
N	4,844
Wind/Hail	
Roof 1	4,354
Roof 2	1,204
Roof 3	644
Theft	
Alarm 1	203
Alarm 2	587
Alarm 3	389
Alarm 4	165

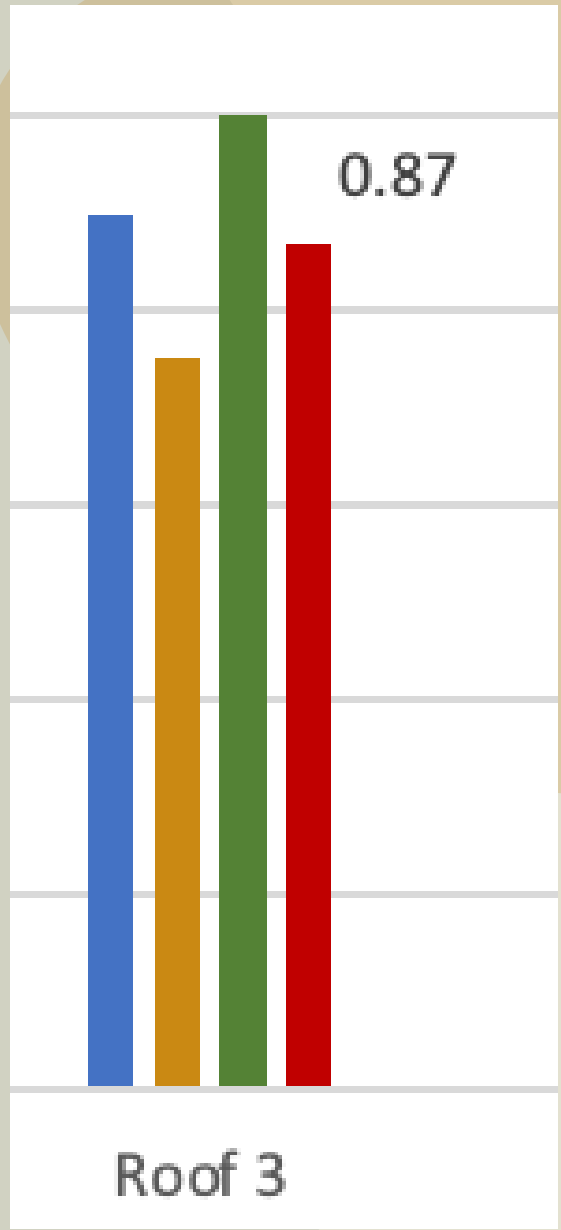
- Information is fully credible only when there are 1,082 claims

Using Competitors Data



Adjustments

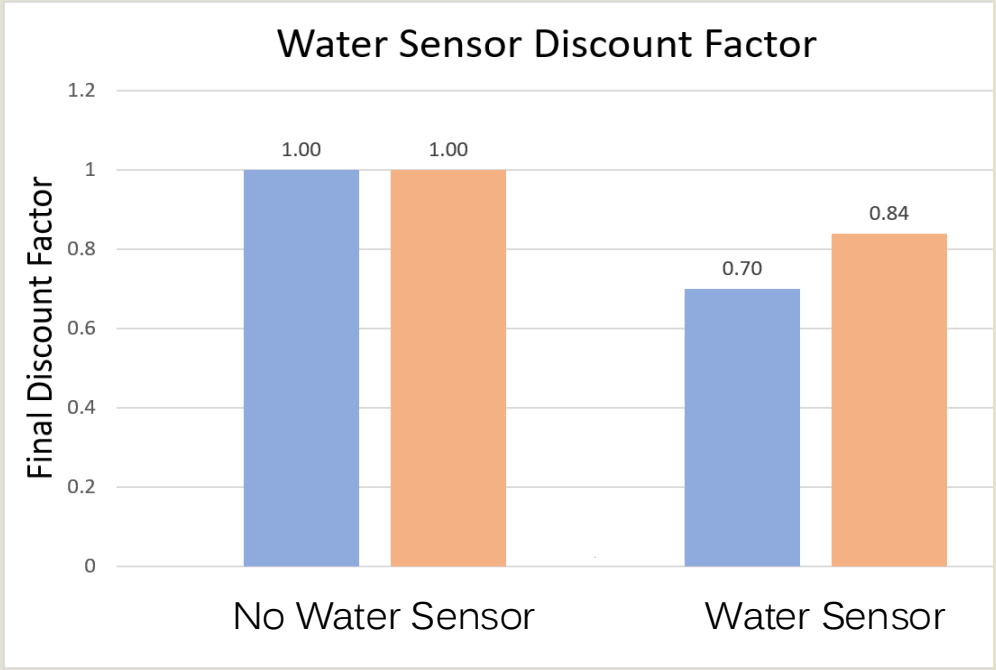
- Roof 3's complement was originally .87
- Changed complement to 1
  - Compared to our competitors, we were discounting too much
  - Roof 3 has solar panels which would require more expensive repairs
- Doing so, we provided a smaller discount

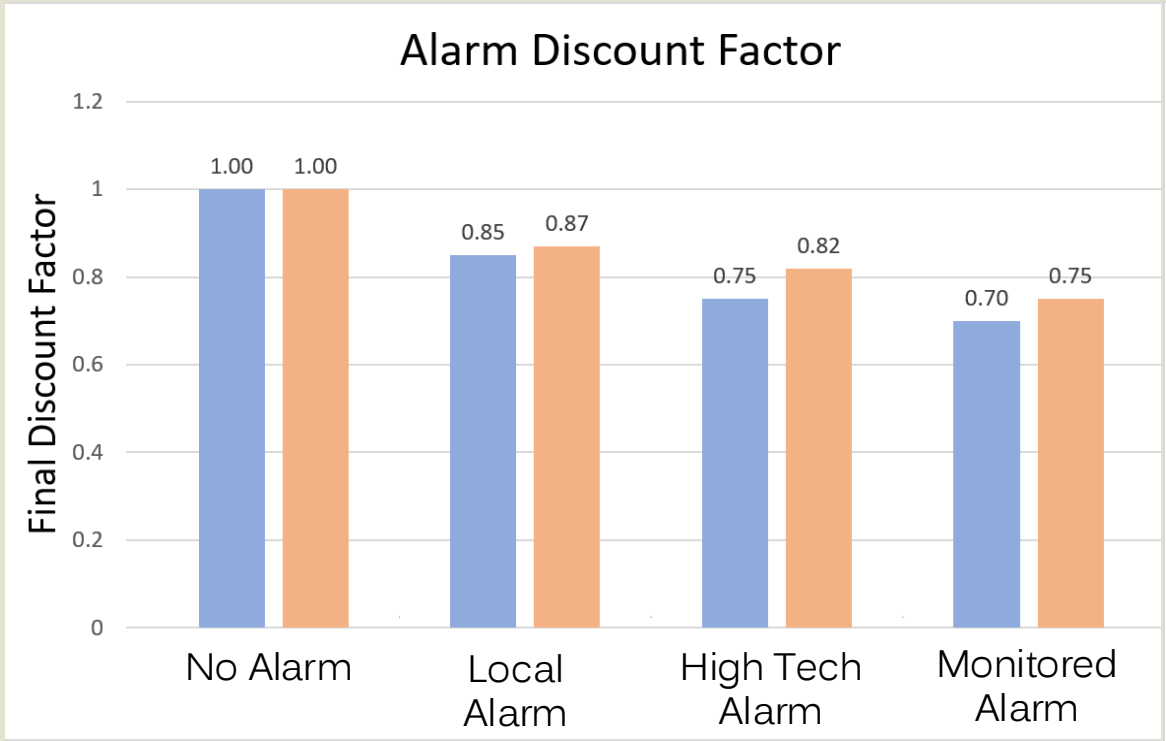


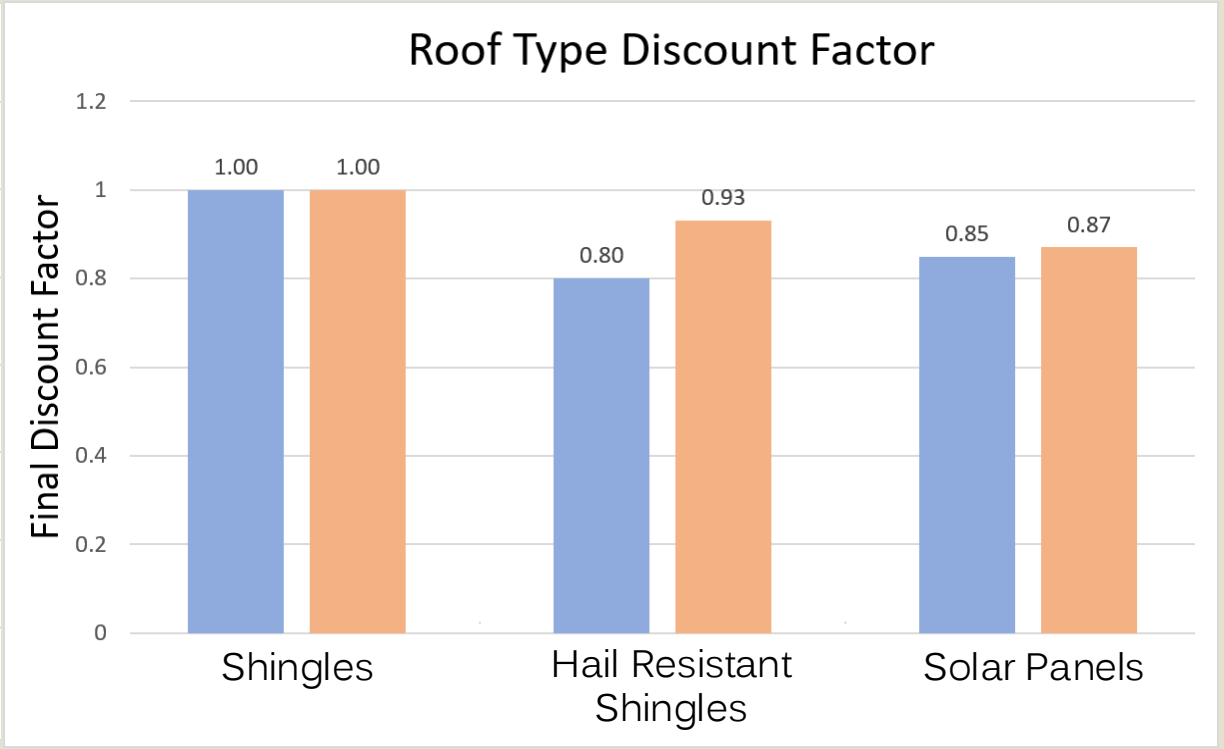
Discount Factors

Alpha Actuary

Average Competitors

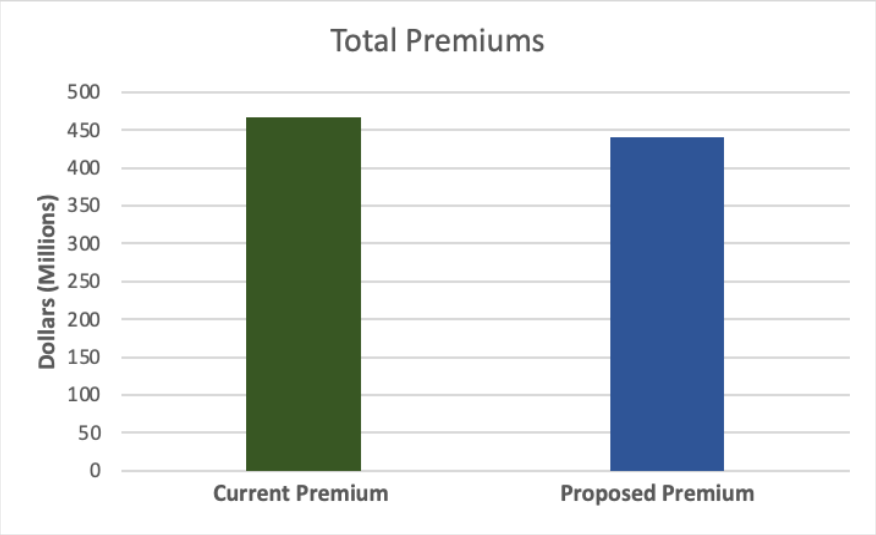






Methodology: Rebasing

Rebasing applies a uniform rate adjustment to **offset** the proposed discounts, **maintaining** the company's overall premium level, while **preserving** discounts for lower-risk policyholders.



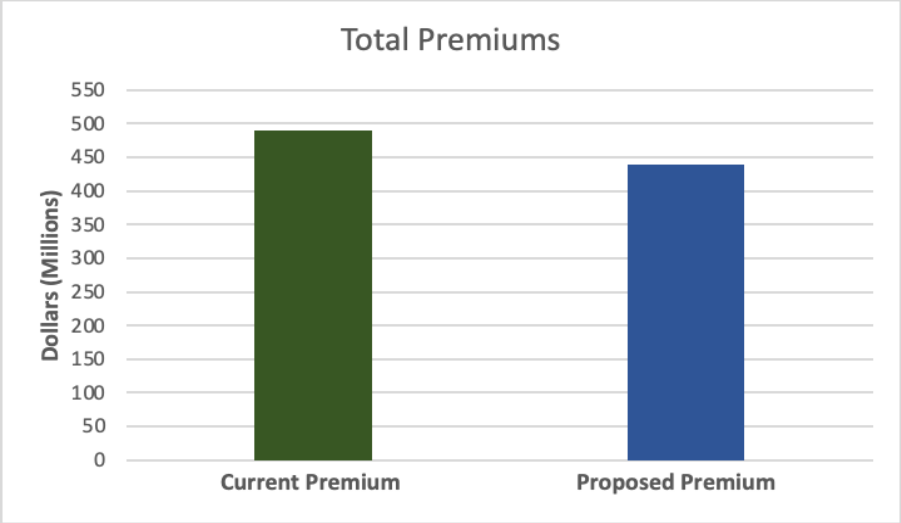
Total Premium Collected in Previous Year:

$466,565,508

Proposed Premium Collected in Previous Year*:

$440,684,722

Hypothetical Graph**

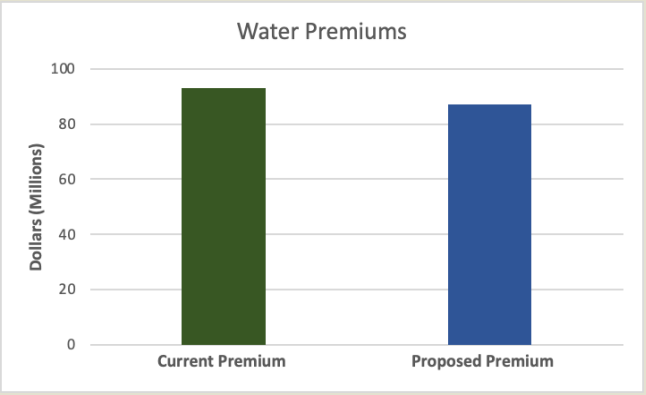


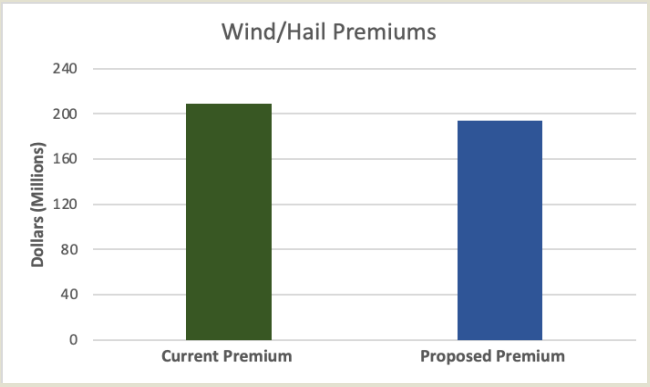
Considering a **Larger** Premium:

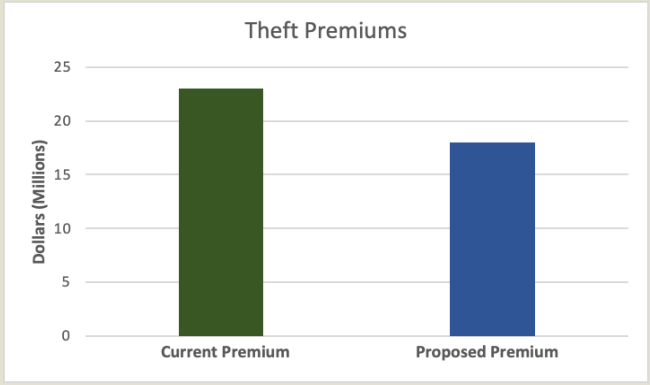
- Current Premium will **increase** while Proposed Premium remains the **same**
- Percentage between Current Premium and Proposed Premium will **increase**
- A larger base rate **increase** is applied to all policies
- Rebalance Factor **increases**

Methodology: Rebasing

Peril	Previous Year Premium	Proposed Premium	Difference	Rebalance Factor
Water	$93.3 M	$87.7 M	-$5.6 M	.0638
Wind/Hail	$209.9 M	$194.1 M	-$15.8 M	.0810
Theft	$23.4 M	$18.8 M	-$4.5 M	.2422

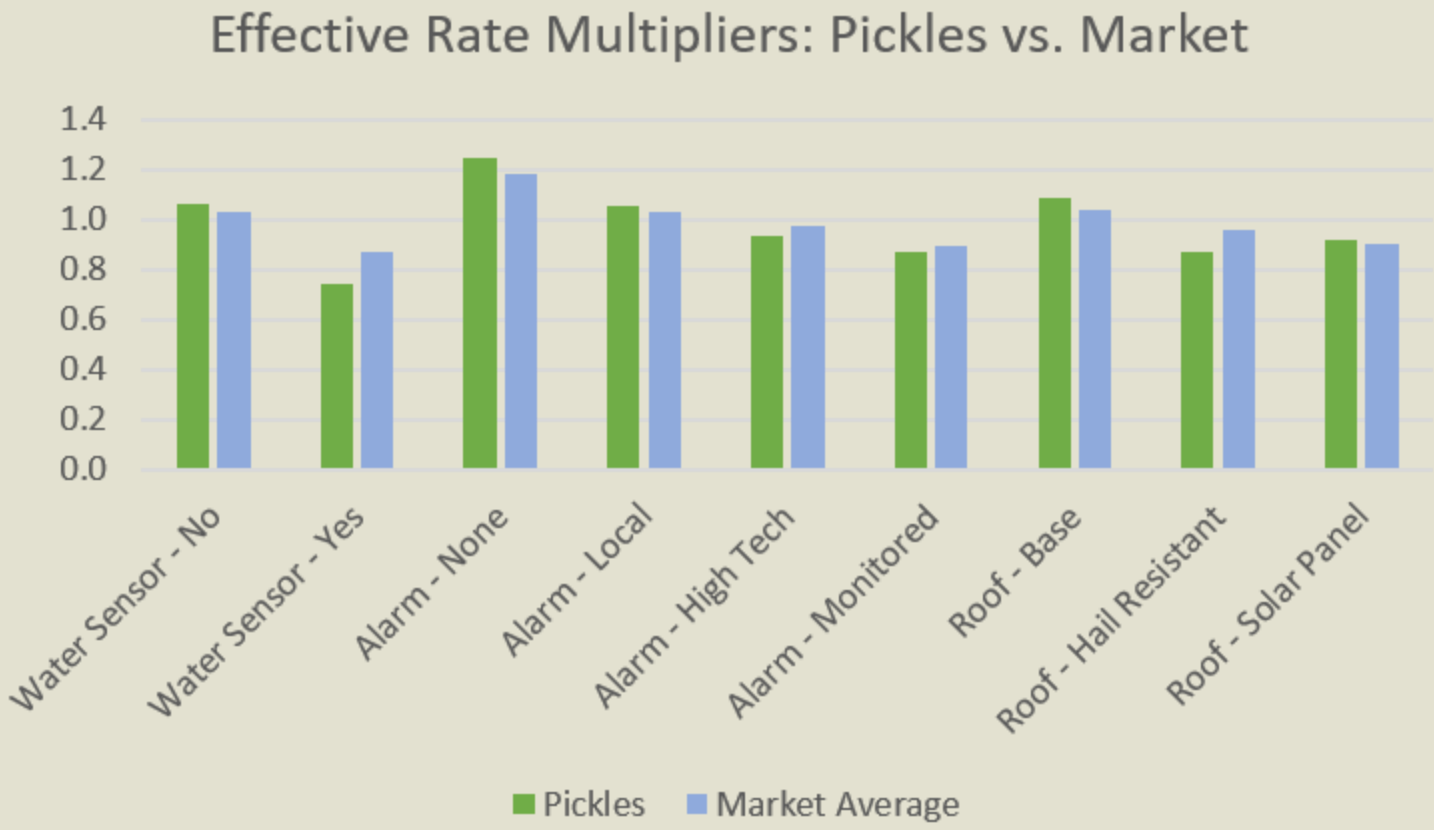






Findings: Market Rate Comparison

- Our data shows our competitors consistently underprice the standard risk while overpricing the mitigated risk
- Our aggressive discount structure allows us to undercut our competition and actively siphon away the most profitable customers
- By raising our base rates to adequate levels, we protect our loss ratio and maintain profitability



Thank You!

Appendix – Loss Ratios (Water Sensor)

Water Sensor (Y/N)						
Premium Collected	Wind/Hail	Water	Fire	Theft	Other	Liability
Y	$42,070,543	$18,668,679	$14,010,281	$4,665,949	$9,336,669	$4,659,713
N	$167,782,039	$74,628,568	$56,056,844	$18,667,590	$37,353,404	$18,665,229
Losses	Wind/Hail	Water	Fire	Theft	Other	Liability
Y	$25,755,146	$7,216,384	$7,229,292	$2,414,748	$6,283,540	$3,118,024
N	$103,338,648	$49,527,502	$28,826,596	$9,595,322	$25,128,438	$12,504,009
Loss ratios	Wind/Hail	Water	Fire	Theft	Other	Liability
Y	61.22%	38.66%	51.60%	51.75%	67.30%	66.91%
N	61.59%	66.37%	51.42%	51.40%	67.27%	66.99%

Appendix – Loss Ratios (Alarm Type)

Alarm Type						
Premium Collected	Wind/Hail	Water	Fire	Theft	Other	Liability
1	$20,990,028	$9,335,974	$6,985,818	$2,334,944	$4,668,425	$2,338,657
2	$83,905,570	$37,259,853	$28,082,399	$9,326,324	$18,661,841	$9,313,911
3	$62,858,969	$28,008,736	$20,969,363	$6,997,825	$14,021,271	$6,990,793
4	$42,098,015	$18,692,684	$14,029,545	$4,674,446	$9,338,536	$4,681,581
Losses	Wind/Hail	Water	Fire	Theft	Other	Liability
1	$12,881,190	$5,997,590	$4,678,189	$1,573,992	$3,142,581	$1,563,129
2	$51,170,238	$22,938,520	$16,043,145	$5,294,912	$12,571,710	$6,241,444
3	$38,968,519	$16,740,304	$9,714,016	$3,265,800	$9,429,988	$4,699,530
4	$26,073,847	$11,067,472	$5,620,538	$1,875,366	$6,267,699	$3,117,930
Loss Ratios	Wind/Hail	Water	Fire	Theft	Other	Liability
1	61.37%	64.24%	66.97%	67.41%	67.32%	66.84%
2	60.99%	61.56%	57.13%	56.77%	67.37%	67.01%
3	61.99%	59.77%	46.32%	46.67%	67.25%	67.22%
4	61.94%	59.21%	40.06%	40.12%	67.12%	66.60%

/2026

Appendix – Loss Ratios (Roof Type)

Roof Type						
Premium Collected	Wind/Hail	Water	Fire	Theft	Other	Liability
1	$125,958,104	$55,957,441	$42,083,342	$13,994,358	$28,061,754	$13,995,482
2	$62,904,433	$27,993,803	$20,982,057	$7,000,734	$13,967,645	$6,994,233
3	$20,990,045	$9,346,003	$7,001,726	$2,338,447	$4,660,674	$2,335,227
Losses	Wind/Hail	Water	Fire	Theft	Other	Liability
1	$84,085,527	$33,988,939	$21,589,161	$7,191,898	$19,060,725	$9,318,914
2	$33,757,546	$17,052,275	$10,854,087	$3,610,186	$9,251,881	$4,746,411
3	$11,250,721	$5,702,672	$3,612,640	$1,207,986	$3,099,372	$1,556,708
Loss Ratios	Wind/Hail	Water	Fire	Theft	Other	Liability
1	66.76%	60.74%	51.30%	51.39%	67.92%	66.59%
2	53.66%	60.91%	51.73%	51.57%	66.24%	67.86%
3	53.60%	61.02%	51.60%	51.66%	66.50%	66.66%

Appendix – Credibility

Credibility Weighted Factor = Z x Indicated Relativity + (1-Z) x Complement

- Indicated Relativity: loss ratio of specific class / loss ratio of baseline class
- Complement: competitors' weighted average by market share
- Partial Credibility/Z: (# of claims / 1082) $^{(1/2)}$

Appendix – Discount Factors

Final Relativity = (Partial Credibility * Indicated Relativity) + (1- Partial Credibility)*Complement

Suggested Discount = 1 – Final Relativity

Round Suggested Discount to Calculate Final Discount

Final Discount Factor = 1 – Final Discount

	Indicated relativity	Complement	Final relativity	**Suggested Discount**	(1- Discount Factor)	**Final Discount**	(1- Final Factor)
Water Sensor			1	--		--	1
Y	0.5825	0.8423	0.6300	**0.3700**	0.6300	**0.3**	0.7
N (base class)							
Alarm							
1 - None (base class)			1	--		--	1
2 - Local Alarm	0.842213495	0.8692	0.84933105	**0.15066895**	0.8493	**0.15**	0.85
3 - High Tech Local Alarm	0.692309762	0.8223	0.744361015	**0.255638985**	0.7444	**0.25**	0.75
4 - Monitored High Tech Alarm	0.595154661	0.7538	0.691876084	**0.308123916**	0.6919	**0.3**	0.7
Roof							
1 - Composite Shingle (base class)			1	--		--	1
2 - Hail Resistant Shingle	0.803886028	N/A	0.803886028	**0.196113972**	0.8039	**0.2**	0.8
3 - Solar Panel over Shingle	0.802919212	**1**	0.847954492	**0.152045508**	0.8480	**0.15**	0.85

Average Competitors Discount Factor was weighted by market share
Summation of (Market Share * Discount Factor)/Summation of Market Share (.65)